Web Programming Lab 4 - Recitation

Question 1

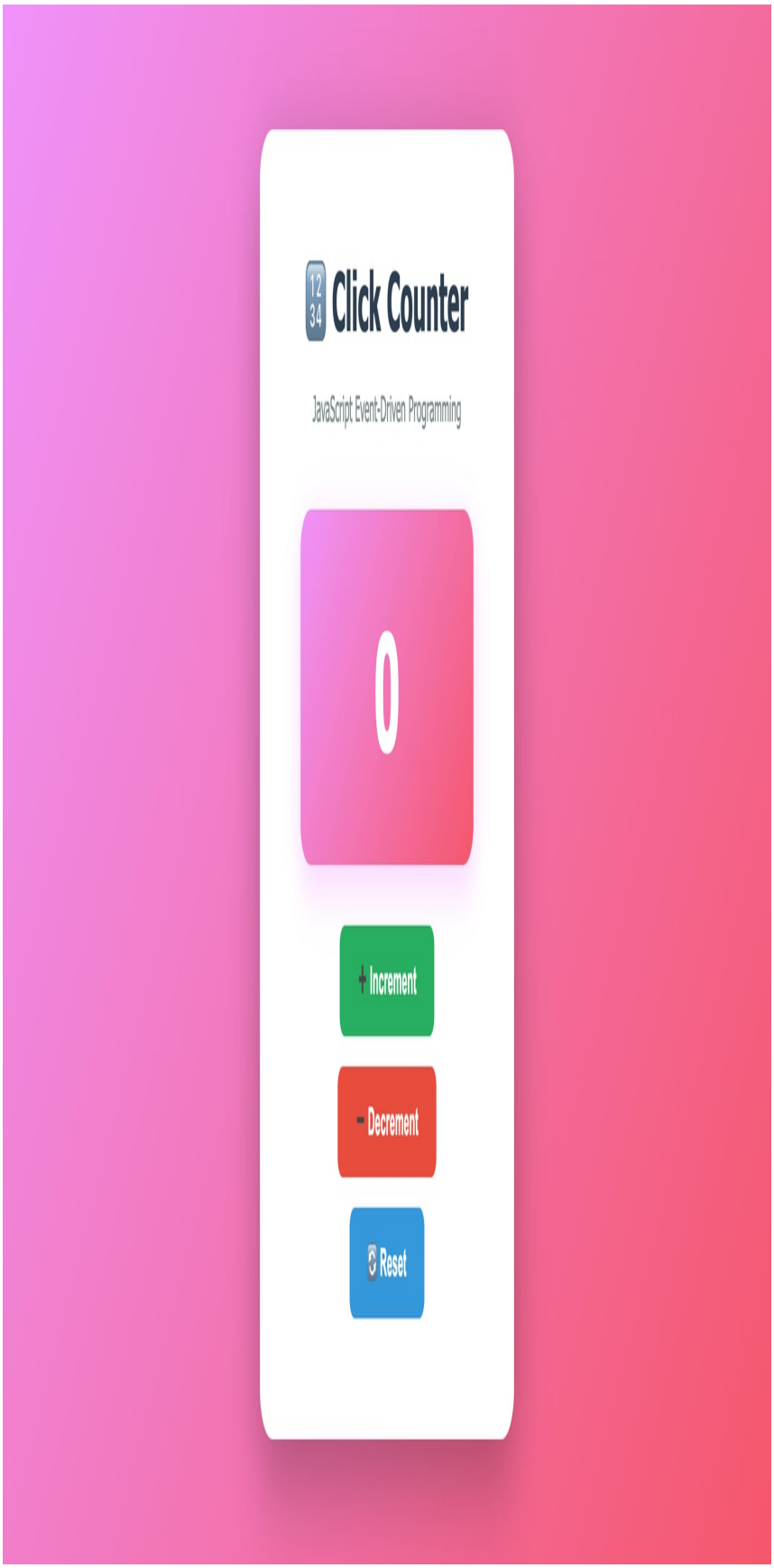
Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Click Counter - JavaScript Events</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%);
            min-height: 100vh;
            display: flex;
            align-items: center;
            justify-content: center;
            padding: 20px;
        }
        .container {
            background: white;
            padding: 60px 80px;
            border-radius: 25px;
            box-shadow: 0 20px 60px rgba(0,0,0,0.3);
            text-align: center;
            max-width: 500px;
            width: 100%;
        }
        h1 {
            font-size: 2.5em;
            color: #2c3e50;
            margin-bottom: 15px;
        }
        .subtitle {
            color: #7f8c8d;
            font-size: 1.1em;
            margin-bottom: 40px;
        }
        .counter-display {
            background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%);
            color: white;
            font-size: 5em;
            font-weight: bold;
            padding: 40px;
            border-radius: 20px;
            margin: 30px 0;
            box-shadow: 0 10px 30px rgba(240, 147, 251, 0.4);
            transition: all 0.3s ease;

        }
        .counter-display.animate {
            transform: scale(1.1);
        }
        .button-group {
            display: flex;
            gap: 15px;
            justify-content: center;
            flex-wrap: wrap;
        }
        button {
            background: #2c3e50;
            color: white;
            border: none;
            padding: 15px 35px;
            font-size: 1.2em;
            border-radius: 10px;
            cursor: pointer;
            transition: all 0.3s ease;
```

```
            font-weight: 600;
        }
        button:hover {
            background: #34495e;
            transform: translateY(-3px);
            box-shadow: 0 8px 20px rgba(0,0,0,0.2);
        }
        button:active {
            transform: translateY(0);
        }
        .increment {
            background: #27ae60;
        }
        .increment:hover {
            background: #229954;
        }
        .decrement {
            background: #e74c3c;
        }
        .decrement:hover {
            background: #c0392b;
        }
        .reset {
            background: #3498db;
        }
        .reset:hover {
            background: #2980b9;
        }
        @media (max-width: 600px) {
            .container {
                padding: 40px 30px;

            }
            .counter-display {
                font-size: 3.5em;
                padding: 30px;
            }
            button {
                padding: 12px 25px;
                font-size: 1em;
            }
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>■ Click Counter</h1>
        <p class="subtitle">JavaScript Event-Driven Programming</p>

        <div class="counter-display" id="counter">0</div>

        <div class="button-group">
            <button class="increment" onclick="increment()">■ Increment</button>
            <button class="decrement" onclick="decrement()">■ Decrement</button>
            <button class="reset" onclick="resetCounter()">■ Reset</button>
        </div>
    </div>

    <script>
        let count = 0;

        function animateCounter() {
            const display = document.getElementById('counter');
            display.classList.add('animate');
            setTimeout(() => display.classList.remove('animate'), 300);
        }

        function increment() {
            count++;
            document.getElementById('counter').textContent = count;
            animateCounter();
        }

        function decrement() {
            count--;
            document.getElementById('counter').textContent = count;
            animateCounter();
        }

        function resetCounter() {
            count = 0;
            document.getElementById('counter').textContent = count;

            animateCounter();
        }
```

```
    </script>
</body>
</html>
```

Output:



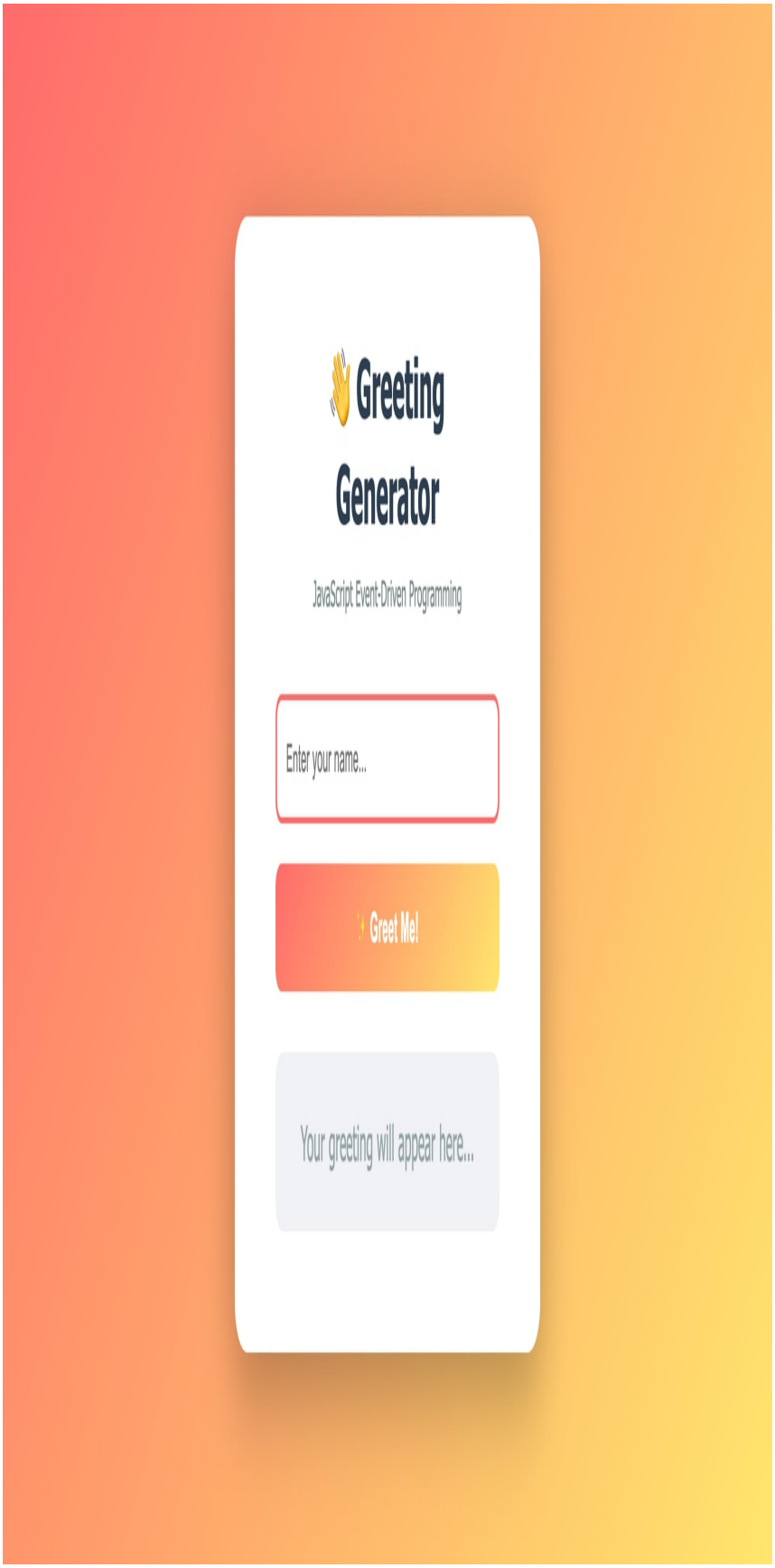
Question 2

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Greeting Generator - JavaScript Events</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #FF6B6B 0%, #FFE66D 100%);
            min-height: 100vh;
            display: flex;
            align-items: center;
            justify-content: center;
            padding: 20px;
        }
        .container {
            background: white;
            padding: 60px 80px;
            border-radius: 25px;
            box-shadow: 0 20px 60px rgba(0,0,0,0.3);
            text-align: center;
            max-width: 600px;
            width: 100%;
        }
        h1 {
            font-size: 2.5em;
            color: #2c3e50;
            margin-bottom: 15px;
        }
        .subtitle {
            color: #7f8c8d;
            font-size: 1.1em;
            margin-bottom: 40px;
        }
        input {
            width: 100%;
            padding: 18px;
            font-size: 1.2em;
            border: 3px solid #FF6B6B;
            border-radius: 12px;
            margin-bottom: 20px;
            transition: all 0.3s ease;
        }
        input:focus {

            outline: none;
            border-color: #FFE66D;
            box-shadow: 0 0 20px rgba(255, 107, 107, 0.3);
        }
        button {
            background: linear-gradient(135deg, #FF6B6B 0%, #FFE66D 100%);
            color: white;
            border: none;
            padding: 18px 50px;
            font-size: 1.3em;
            border-radius: 12px;
            cursor: pointer;
            transition: all 0.3s ease;
            font-weight: 600;
            width: 100%;
        }
        button:hover {
            transform: translateY(-3px);
            box-shadow: 0 10px 30px rgba(255, 107, 107, 0.4);
        }
        button:active {
            transform: translateY(0);
        }
        #greetOutput {
            margin-top: 30px;
            padding: 30px;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
```

```
            color: white;
            border-radius: 15px;
            font-size: 1.5em;
            min-height: 80px;
            display: flex;
            align-items: center;
            justify-content: center;
            box-shadow: 0 10px 30px rgba(102, 126, 234, 0.3);
            transition: all 0.3s ease;
        }
        #greetOutput:empty {
            background: #f0f2f5;
            box-shadow: none;
        }
        #greetOutput:empty:before {
            content: 'Your greeting will appear here...';
            color: #95a5a6;
            font-size: 1em;
        }
        .emoji {
            font-size: 2em;
            margin-right: 15px;
        }

        @media (max-width: 600px) {
            .container {
                padding: 40px 30px;
            }
            h1 {
                font-size: 2em;
            }
            button, input {
                font-size: 1em;
                padding: 15px;
            }
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>■ Greeting Generator</h1>
        <p class="subtitle">JavaScript Event-Driven Programming</p>

        <input type="text" id="nameInput" placeholder="Enter your name..." autocomplete="off">
        <button onclick="greet()">■ Greet Me!</button>

        <div id="greetOutput"></div>
    </div>

    <script>
        function greet() {
            const name = document.getElementById('nameInput').value.trim();
            const hour = new Date().getHours();

            let greeting, emoji;
            if (hour < 12) {
                greeting = 'Good Morning';
                emoji = '■';
            } else if (hour < 17) {
                greeting = 'Good Afternoon';
                emoji = '■■';
            } else {
                greeting = 'Good Evening';
                emoji = '■';
            }

            const output = document.getElementById('greetOutput');

            if (name) {
                output.innerHTML = `<span class="emoji">${emoji}</span><div><strong>${greeting}, ${name}!</strong
            } else {
                output.innerHTML = `<div>Please enter your name! ■</div>`;
            }
        }

        // Allow Enter key to submit
        document.getElementById('nameInput').addEventListener('keypress', function(event) {
            if (event.key === 'Enter') {
                greet();
            }
        });
    </script>
</body>
</html>
```

Output:


Greeting
Generator
JavaScript Event-Driven Programming
Enter your name...
Greet Me!
Your greeting will appear here...

Question 3

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Simple Calculator - JavaScript Events</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #11998e 0%, #38ef7d 100%);
            min-height: 100vh;
            display: flex;
            align-items: center;
            justify-content: center;
            padding: 20px;
        }
        .container {
            background: white;
            padding: 60px 80px;
            border-radius: 25px;
            box-shadow: 0 20px 60px rgba(0,0,0,0.3);
            text-align: center;
            max-width: 550px;
            width: 100%;
        }
        h1 {
            font-size: 2.5em;
            color: #2c3e50;
            margin-bottom: 15px;
        }
        .subtitle {
            color: #7f8c8d;
            font-size: 1.1em;
            margin-bottom: 40px;
        }
        .calc-row {
            display: flex;
            gap: 15px;
            margin-bottom: 25px;
            align-items: center;
            justify-content: center;
        }
        input[type="number"] {
            width: 150px;
            padding: 18px;

            font-size: 1.3em;
            border: 3px solid #11998e;
            border-radius: 12px;
            text-align: center;
            transition: all 0.3s ease;
        }
        input[type="number"]:focus {
            outline: none;
            border-color: #38ef7d;
            box-shadow: 0 0 20px rgba(17, 153, 142, 0.3);
        }
        select {
            padding: 18px 15px;
            font-size: 1.5em;
            border: 3px solid #11998e;
            border-radius: 12px;
            background: white;
            cursor: pointer;
            transition: all 0.3s ease;
        }
        select:focus {
            outline: none;
            border-color: #38ef7d;
        }
        button {
            background: linear-gradient(135deg, #11998e 0%, #38ef7d 100%);
            color: white;
```

```
            border: none;
            padding: 18px 50px;
            font-size: 1.3em;
            border-radius: 12px;
            cursor: pointer;
            transition: all 0.3s ease;
            font-weight: 600;
            width: 100%;
            margin-top: 10px;
        }
        button:hover {
            transform: translateY(-3px);
            box-shadow: 0 10px 30px rgba(17, 153, 142, 0.4);
        }
        button:active {
            transform: translateY(0);
        }
        #result {
            margin-top: 30px;
            padding: 30px;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            color: white;
            border-radius: 15px;

            font-size: 2.5em;
            font-weight: bold;
            min-height: 100px;
            display: flex;
            align-items: center;
            justify-content: center;
            box-shadow: 0 10px 30px rgba(102, 126, 234, 0.3);
            transition: all 0.3s ease;
        }
        #result:empty {
            background: #f0f2f5;
            box-shadow: none;
        }
        #result:empty:before {
            content: '= ?';
            color: #95a5a6;
            font-size: 0.6em;
        }
        @media (max-width: 600px) {
            .container {
                padding: 40px 30px;
            }
            .calc-row {
                flex-direction: column;
            }
            input[type="number"] {
                width: 100%;
            }
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>■ Simple Calculator</h1>
        <p class="subtitle">JavaScript Event-Driven Programming</p>

        <div class="calc-row">
            <input type="number" id="num1" placeholder="Number 1" value="">
            <select id="operator">
                <option value="+">+</option>
                <option value="-">−</option>
                <option value="*">×</option>
                <option value="/">÷</option>
            </select>
            <input type="number" id="num2" placeholder="Number 2" value="">
        </div>

        <button onclick="calculate()">■ Calculate</button>

        <div id="result"></div>

    </div>

    <script>
        function calculate() {
            const num1 = parseFloat(document.getElementById('num1').value);
            const num2 = parseFloat(document.getElementById('num2').value);
            const operator = document.getElementById('operator').value;
            const resultDiv = document.getElementById('result');

            if (isNaN(num1) || isNaN(num2)) {
```

```
                resultDiv.textContent = 'Please enter valid numbers!';
                resultDiv.style.background = 'linear-gradient(135deg, #e74c3c 0%, #c0392b 100%)';
                return;
            }

            let result;
            switch(operator) {
                case '+': result = num1 + num2; break;
                case '-': result = num1 - num2; break;
                case '*': result = num1 * num2; break;
                case '/':
                    if (num2 === 0) {
                        resultDiv.textContent = 'Cannot divide by zero!';
                        resultDiv.style.background = 'linear-gradient(135deg, #e74c3c 0%, #c0392b 100%)';
                        return;
                    }
                    result = num1 / num2;
                    break;
            }

            result = Math.round(result * 100000) / 100000; // Round to 5 decimals
            resultDiv.textContent = `= ${result}`;
            resultDiv.style.background = 'linear-gradient(135deg, #667eea 0%, #764ba2 100%)';
        }

        // Allow Enter key to calculate
        document.querySelectorAll('input[type="number"]').forEach(input => {
            input.addEventListener('keypress', function(event) {
                if (event.key === 'Enter') {
                    calculate();
                }
            });
        });
    </script>
</body>
</html>
```

Output:

Simple Calculator

JavaScript Event-Driven Programming

Number 1 | + | Number 2

Calculate

=?

Question 4

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Color Changer - JavaScript Events</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #2C3E50 0%, #4CA1AF 100%);
            min-height: 100vh;
            display: flex;
            align-items: center;
            justify-content: center;
            padding: 20px;
        }
        .container {
            background: white;
            padding: 60px 80px;
            border-radius: 25px;
            box-shadow: 0 20px 60px rgba(0,0,0,0.3);
            text-align: center;
            max-width: 600px;
            width: 100%;
        }
        h1 {
            font-size: 2.5em;
            color: #2c3e50;
            margin-bottom: 15px;
        }
        .subtitle {
            color: #7f8c8d;
            font-size: 1.1em;
            margin-bottom: 40px;
        }
        .color-box {
            width: 100%;
            height: 300px;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            border-radius: 20px;
            display: flex;
            align-items: center;
            justify-content: center;
            font-size: 2em;
            color: white;

            font-weight: bold;
            cursor: pointer;
            transition: all 0.4s ease;
            box-shadow: 0 10px 30px rgba(0,0,0,0.2);
            margin-bottom: 30px;
        }
        .color-box:hover {
            transform: scale(1.05) rotate(2deg);
            box-shadow: 0 20px 50px rgba(0,0,0,0.3);
        }
        .color-box:active {
            transform: scale(0.98);
        }
        .button-group {
            display: flex;
            gap: 15px;
            flex-wrap: wrap;
        }
        button {
            flex: 1;
            min-width: 150px;
            background: linear-gradient(135deg, #2C3E50 0%, #4CA1AF 100%);
            color: white;
            border: none;
            padding: 18px 30px;
            font-size: 1.2em;
            border-radius: 12px;
```

```
            cursor: pointer;
            transition: all 0.3s ease;
            font-weight: 600;
        }
        button:hover {
            transform: translateY(-3px);
            box-shadow: 0 10px 30px rgba(44, 62, 80, 0.4);
        }
        button:active {
            transform: translateY(0);
        }
        #colorCode {
            margin-top: 20px;
            padding: 15px;
            background: #f0f2f5;
            border-radius: 10px;
            font-size: 1.3em;
            font-weight: bold;
            color: #2c3e50;
        }
        @media (max-width: 600px) {
            .container {
                padding: 40px 30px;

            }
            .color-box {
                height: 250px;
                font-size: 1.5em;
            }
            button {
                width: 100%;
            }
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>■ Color Changer</h1>
        <p class="subtitle">JavaScript Event-Driven Programming</p>

        <div class="color-box" id="colorBox"
             onmouseover="showHoverEffect()"
             onmouseout="hideHoverEffect()"
             onclick="changeColor()">
            Hover & Click Me! ■■
        </div>

        <div class="button-group">
            <button onclick="changeColor()">■ Random Color</button>
            <button onclick="resetColor()">■ Reset</button>
        </div>

        <div id="colorCode">#667eea → #764ba2</div>
    </div>

    <script>
        const defaultGradient = 'linear-gradient(135deg, #667eea 0%, #764ba2 100%)';
        let currentColor = '#667eea';

        function getRandomColor() {
            return '#' + Math.floor(Math.random() * 16777215).toString(16).padStart(6, '0');
        }

        function changeColor() {
            const color1 = getRandomColor();
            const color2 = getRandomColor();
            const gradient = `linear-gradient(135deg, ${color1} 0%, ${color2} 100%)`;

            const box = document.getElementById('colorBox');
            box.style.background = gradient;
            currentColor = color1;

            document.getElementById('colorCode').textContent = `${color1} → ${color2}`;
        }

        function resetColor() {
            const box = document.getElementById('colorBox');
            box.style.background = defaultGradient;
            document.getElementById('colorCode').textContent = '#667eea → #764ba2';
        }

        function showHoverEffect() {
            const box = document.getElementById('colorBox');
            box.textContent = '■ Click for Random Colors!';
        }
```

```
        function hideHoverEffect() {
            const box = document.getElementById('colorBox');
            box.textContent = 'Hover & Click Me! ■■';
        }
    </script>
</body>
</html>
```

Output:

🎨 Color Changer
JavaScript Event-Driven Programming
Hover & Click Me!
🔀 Random Color
🔄 Reset
#667eea → #764ba2

Question 5

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Event Log - JavaScript Events</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #8E2DE2 0%, #4A00E0 100%);
            min-height: 100vh;
            display: flex;
            align-items: center;
            justify-content: center;
            padding: 20px;
        }
        .container {
            background: white;
            padding: 60px 80px;
            border-radius: 25px;
            box-shadow: 0 20px 60px rgba(0,0,0,0.3);
            text-align: center;
            max-width: 700px;
            width: 100%;
        }
        h1 {
            font-size: 2.5em;
            color: #2c3e50;
            margin-bottom: 15px;
        }
        .subtitle {
            color: #7f8c8d;
            font-size: 1.1em;
            margin-bottom: 40px;
        }
        input {
            width: 100%;
            padding: 20px;
            font-size: 1.3em;
            border: 3px solid #8E2DE2;
            border-radius: 12px;
            margin-bottom: 30px;
            transition: all 0.3s ease;
        }
        input:focus {

            outline: none;
            border-color: #4A00E0;
            box-shadow: 0 0 20px rgba(142, 45, 226, 0.3);
        }
        #output {
            background: #f8f9fa;
            border-radius: 15px;
            padding: 20px;
            max-height: 400px;
            overflow-y: auto;
            text-align: left;
        }
        .event-item {
            background: white;
            padding: 15px 20px;
            margin: 10px 0;
            border-left: 5px solid #8E2DE2;
            border-radius: 8px;
            box-shadow: 0 2px 8px rgba(0,0,0,0.08);
            animation: slideIn 0.3s ease;
        }
        @keyframes slideIn {
            from {
                opacity: 0;
                transform: translateX(-20px);
            }
            to {
```

```
                opacity: 1;
                transform: translateX(0);
            }
        }
        .event-time {
            color: #8E2DE2;
            font-weight: bold;
            font-size: 0.9em;
            margin-right: 10px;
        }
        .event-key {
            display: inline-block;
            background: linear-gradient(135deg, #8E2DE2 0%, #4A00E0 100%);
            color: white;
            padding: 5px 12px;
            border-radius: 6px;
            font-weight: bold;
            margin: 0 5px;
        }
        .event-type {
            color: #7f8c8d;
            font-size: 0.85em;
            margin-top: 5px;

        }
        .clear-btn {
            background: linear-gradient(135deg, #e74c3c 0%, #c0392b 100%);
            color: white;
            border: none;
            padding: 12px 30px;
            font-size: 1em;
            border-radius: 8px;
            cursor: pointer;
            transition: all 0.3s ease;
            font-weight: 600;
            margin-top: 20px;
        }
        .clear-btn:hover {
            transform: translateY(-2px);
            box-shadow: 0 8px 20px rgba(231, 76, 60, 0.3);
        }
        #output:empty:before {
            content: '■■ Start typing to see keyboard events...';
            color: #95a5a6;
            display: flex;
            align-items: center;
            justify-content: center;
            height: 200px;
            font-size: 1.1em;
        }
        @media (max-width: 600px) {
            .container {
                padding: 40px 30px;
            }
            h1 {
                font-size: 2em;
            }
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>■■ Event Log</h1>
        <p class="subtitle">JavaScript Event-Driven Programming</p>

        <input type="text"
               id="eventInput"
               placeholder="Type here to trigger events..."
               onkeyup="logEvent(event)"
               autocomplete="off">

        <div id="output"></div>

        <button class="clear-btn" onclick="clearLog()">■■ Clear Log</button>

    </div>

    <script>
        let eventCount = 0;

        function logEvent(event) {
            eventCount++;
            const output = document.getElementById('output');
            const time = new Date().toLocaleTimeString();
```

```
            const eventItem = document.createElement('div');
            eventItem.className = 'event-item';

            let keyDisplay = event.key;
            if (event.key === ' ') keyDisplay = 'Space';
            if (event.key === 'Enter') keyDisplay = 'Enter';
            if (event.key === 'Backspace') keyDisplay = 'Backspace';

            eventItem.innerHTML = `
                <div>
                    <span class="event-time">${time}</span>
                    Event #${eventCount}: Key pressed →
                    <span class="event-key">${keyDisplay}</span>
                </div>
                <div class="event-type">
                    Type: ${event.type} | Code: ${event.code} | Key: ${event.key}
                </div>
            `;

            output.insertBefore(eventItem, output.firstChild);

            // Limit to 20 events
            while (output.children.length > 20) {
                output.removeChild(output.lastChild);
            }
        }

        function clearLog() {
            document.getElementById('output').innerHTML = '';
            document.getElementById('eventInput').value = '';
            eventCount = 0;
        }
    </script>
</body>
</html>
```

Output:

Event Log

JavaScript Event-Driven Programming

Type here to trigger events...

Start typing to see keyboard events...

Clear Log